

COPPER TAPE LIGHTHOUSE



Light Up Your Project!

STEM & Electronics for Grade 1

THE STORY OF THE SEA

Why do ships need lights?

Imagine standing on a beach at night. The sea is **dark and dangerous**! Ships out in the ocean need a way to find their way safely back to the shore. [cite: 288]

Lighthouses: The Tall Guides

Lighthouses are tall towers that shine a bright beam of light across the water. In India, **Coastal Guards** use these towers to help guide boats through the darkness. [cite: 289]

Today, we are going to build our own lighthouse using a "**Path of Electricity**"! [cite: 289]

Let's turn on the lights!



Meet the Tools

To make electricity flow, we need a few "Circuit Friends." Let's learn their names: [cite: 290, 291]



LED - The "Small Light"

This is a tiny lamp. It is very special because it only lights up when electricity flows through it the right way! [cite: 292]



Coin Cell Battery - The "Power Button"

This little silver disc holds all the energy. It is the heart of our circuit! [cite: 292, 296]



Copper Tape - The "Shiny Tracks"

Electricity loves to travel on copper. We will use this sticky tape to build a road for the energy to follow. [cite: 292]

What is a Circuit?

Electricity flows just like **water in a pipe**! [cite: 294]

A circuit is a **Path** that must be "**Bandh**" (**Closed**) for electricity to move. [cite: 294]

Think of it as a **Loop**:

- It starts at the Battery. [cite: 295]
- Travels through the Shiny Copper Tape. [cite: 295]
- Lights up the LED. [cite: 295]
- Goes all the way back to the Battery! [cite: 295]

[Image of a simple paper circuit loop with a battery and LED]

If the loop is broken, the light goes out!

INDIAN LIGHTHOUSES

Amazing Coastal Towers

India has many famous lighthouses that protect our coastline. Have you heard of these? [cite: 303, 304]

Dolphin's Nose

Located in **Visakhapatnam**, this lighthouse sits high on a hill that looks like a dolphin's nose! [cite: 304]

Aguada Lighthouse

A very old and famous lighthouse in **Goa** that has watched over ships for many years. [cite: 304]

Did you know? Many lighthouses in **Gujarat** and **Kerala** are painted with bold **Red**-and-White stripes! [cite: 305]

Building the Track

It's time to become an **Electricity Detective!** [cite: 311]

Step 1: Peel and Stick

Stick your copper tape along the dotted lines on your lighthouse template. [cite: 307]

Step 2: Press Firmly

Use your fingers to press down hard. Make sure your "Shiny Track" is continuous with **no gaps or breaks!** [cite: 307]

Step 3: Connect the Friends

Place your coin battery on the tape (Positive + side up!). [cite: 309]

Put the LED legs on the tracks: **Long leg** on Positive (+), **Short leg** on Negative (-). [cite: 310]

CHAMATKAR! (MAGIC)

Press down gently and watch what happens...



Detective Challenge!

Light not glowing? Check your path: [cite: 318, 319]

- Are there any gaps in the copper tape? [cite: 320]
- Is the battery flat side up? [cite: 320]
- Try flipping the LED around! [cite: 321]

Success Criteria:

- ✓ I can identify the battery, tape, and LED. [cite: 327]
- ✓ I know electricity needs a complete loop. [cite: 327]
- ✓ My lighthouse LED glows bright! [cite: 315, 327]